will be positive.

(b) If the displacement is in a direction opposite to the force, i.e., $\theta = 180°$ then $\cos\theta = -1$ and work will be negative.

(c) When the displacement is normal to the direction of force, i.e., $\theta = 90°$ then $\cos\theta = 0$ and ~~word work~~ work done will be 0.

(ii) The SI unit and CGS unit of force are N (newton) and Dyne. The SI and CGS units of work are Joule and dyne-cm (erg).

(iii) taking g as $10\ m/s^2$

(a) force against gravity $= 40 \times 10 = 400N$

(b) $W = F \times S$

$W = 400 \times 8$

$W = 3200J$

(c) $P = \frac{W}{t}$

$P = \frac{3200}{5}$

$P = 640W$

(ii) $VR = 4$

$\eta = 90\%$

(a) $MA = VR \times \eta$

$MA = \frac{4 \times 90}{100} = 3.6$

(b) $MA = \frac{L}{E}$

$E = \frac{L}{MA}$

$E = \frac{300}{3.6}$

$E = 83.33N$

(iii)

(a) The ratio of the velocity of effort to the velocity of the load is called velocity ratio.

(b) The ratio of the load to the effort is called mechanical advantage.

(c) An ideal machine is that in which there is no loss in energy in any manner.

(d) The machine is said to be an ideal machine when the work output is equal to the work input.

6

(i)

(a) If the displacement is in the direction of the force, i.e., $\theta = 0°$, then $\cos\theta° = 1$ and work system. Examples- friction.

(ii) $P = \frac{W}{t}$

$W = f \times s$

$f = 20000 \times 9.8 = 196000N$

196000×20

$3920000 J$

$P = \frac{3920000}{60} = 65333.34 W$

(iii)

(a) The potential energy possesed by ~~by~~ a body due to the force of attraction of Earth on it is called as gravitational potential energy.

(b) $m = 2.5 kg$

$g = 10 m/s^2$

$h = 15m$

$U = mgh$

$U = 10 \times 2.5 \times 15 = 375 J$

5

(i) $MA = VR \times \eta$

mechanical advantage = velocity ratio × efficiency

Efficiency of a machine is the ratio of the work done ~~on the load by the machine to the work done~~ on the load by the machine to the work done on the machine by the effort.

(ii) $A = 60^\circ$

refractive index = 1.5

$r = \frac{A}{2} = 30^\circ$

Using Snells law,

$n_{air} \sin i = n_{glass} \sin r$

$\sin i = 1.5 \sin 30^\circ$

$i = 48.6^\circ$

$\delta = 2i - A$

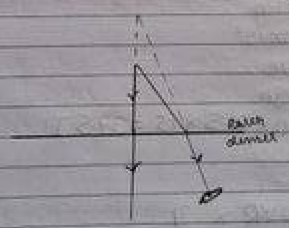
$\delta = 2(48.6) - 60^\circ$

$\delta = 37.9^\circ$

(iii) Two optical illusions based on refraction of light are:-

→ The point on paper appears to be raised when a glass slab is placed over it.

→ A water tank appears shallower than it actually is.



4

(i) Conservative forces are forces that depend only on the initial and final positions of an object's motion and conserve the total energy of the system. Example - gravitational force.

Non-conservative forces are forces that depend on the path or velocity of an object's motion,

(iii) Apparent depth = $\frac{\text{real depth}}{{}^{a}\mu_{m}}$

(iv) $\cancel{MA = \frac{L}{E}}$ $\quad L = T + T$

$L = 2T$

$\cancel{MA = \frac{2T}{T}}$ $\quad E = T$

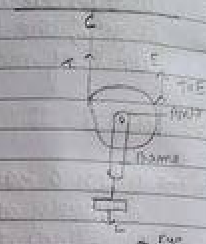
$E = \frac{L}{2}$

$\cancel{MA = 2}$

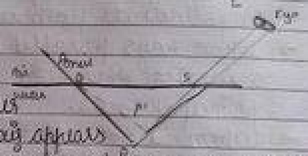
$MA = \frac{L}{E}$

$\cancel{L = T + T}$

$MA = \frac{2T}{T} = 2$

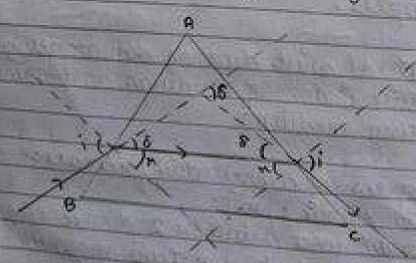


(v) Due to refraction of light, the portion of pencil under the water when seen in air appears to be shortened and raised and makes the pencil look bent.



Section - B

3 (i) The angle of deviation is the angle between the direction of incident ray and the direction of the refracted or emergent ray.



(iv) The principle of moments states that if the algebraic sum of moments of all the forces, acting on the body, about the axis of rotation is zero, the body is in equilibrium. The CGS unit of moment of force is Dyne - cm.

(v) 420 J

(vi) → When a ray of light strikes the convex lens obliquely at its optical centre, it continues to follow its path without any deviation.

→ When rays parallel to the principal axis strikes a convex lens, the rays are converged to the principal focus.

→ When a ray passing through the principal focus strikes a convex lens, the refracted ray will be parallel to the principal axis.

(vii) The deviation of the incident light rays produced by a lens on refraction through it is a measure of its power. It's SI unit is dioptre.

17.(i) The lateral shift depends on angle of incidence, refractive index of medium and thickness of refracting medium.

(ii) Examples of conservative forces are:-

→ Force in elastic spring

→ Electrostatic force between 2 electric charges.

→ Magnetic force between two magnetic poles.

15 (4) Convex lens

16 (i) According to work energy theorem, the increase in kinetic energy of a moving body is equal to the work done by a force acting in the direction of the moving body.

$\frac{1}{2} m(v^2 - u^2)$

$\frac{1}{2} \times 1(100 - 25)$

$\frac{1}{2} \times 75 = 37.5 J$

(ii) → Focal length of convex lens is positive.
→ Focal length of concave lens is negative.
→ Distances measured from the lens's optical centre to the right side are considered positive.
→ Distances measured in the opposite direction of incident ray are negative.
→ Positive distances are those measured in the same direction as the incident ray.
→ Positive heights are those measured ~~downwards~~ ~~and~~ upwards and perpendicular to principal axis
→ Negative heights are those measured in the ~~same~~ ~~direction~~ downwards and perpendicular to principal axis

(iii) The ratio of the length of image I perpendicular to the principal axis, to the length of object O, is called linear magnification.

Physics

Section A

1 (3) 100 %

2 (3) Zero

$g = 0$
displacement $= 0$
$w = 0$

3 (2) Class II lever

4 (2) 2

5 (2) Total mechanical energy

6 (1) greater than 20 cm

7 (4) All of the above

8 (1) 50 N-m clockwise

9 (2) 980

10 (2) Vector quantity

11 (4) The body ~~which~~ will have rotational as well as translational motion

12 (3) Friction force

13 (4) Zero

14 (1) 6 m/s

PE = KE + loss
10% = 0.10 PE
PE = KE + 0.1 PE = 0.9 PE = KE
$0.9 mgh = 0.5 v^2 \Rightarrow v^2 = 1.8 gh$